

MATH 8090: Autoregressive Integrated Moving Average (ARIMA) Models and Seasonal ARIMA Models

Readings: CC08 Chapters 5.1–5.3, 6.4, and 10; BD16 Chapters 6.1–6.2 and 6.4–6.5; SS17 Chapters 3.6–3.7 and 3.9

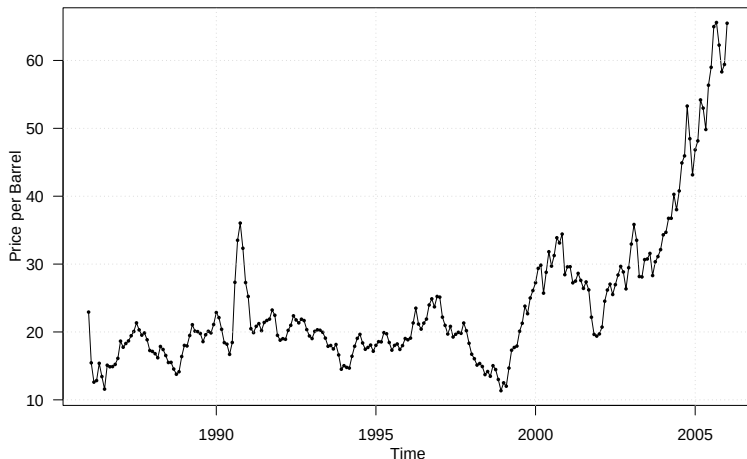
Whitney Huang



Week 8

Autoregressive Integrated Moving Average (ARIMA) Models

Monthly Price of Oil: January 1986–January 2006



A stationary model does not seem to be reasonable. However, it is also not clear which (deterministic) trend model is appropriate 😞

Random Walks Revisited

Recall the random walk process

$$X_t = Z_1 + Z_2 + \cdots + Z_t = \sum_{j=1}^t Z_j,$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$

$\{X_t\}$ is a **nonstationary process**

- We can obtain a **stationary** process by **differencing**

$$\nabla X_t = X_t - X_{t-1} = (1 - B)X_t = Z_t$$

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- We can obtain a **stationary** process by **differencing**

$$\nabla X_t = X_t - X_{t-1} = (1 - B)X_t = Z_t$$

- $\{X_t\}$ is an example of an **autoregressive integrated moving average** (ARIMA) process— ARIMA(0, 1, 0) process

ARIMA Models

Key idea: An ARIMA model describes a nonstationary series X_t whose d th difference is a stationary ARMA process.

- ▶ Let d be a non-negative integer. Then X_t is an ARIMA(p, d, q) process if

$$Y_t = \nabla^d X_t = (1 - B)^d X_t$$

is a **causal** ARMA process

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- ▶ Let $\phi(B)$ be the AR polynomial and $\theta(B)$ be the MA polynomial. Then for $\{Z_t\} \sim \text{WN}(0, \sigma^2)$

$$\phi(B)Y_t = \theta(B)Z_t,$$

and since $Y_t = (1 - B)^d X_t$, we have

$$\phi(B)(1 - B)^d X_t = \theta(B)Z_t$$

Example: ARIMA(1, 1, 0)

Let $\phi(z) = 1 - \phi_1 z$, $\theta(z) = 1$ and $d = 1$. For a **causal stationary solution** (after differencing) we need to assume $|\phi_1| < 1$. Then $\{X_t\}$ is an ARIMA (1, 1, 0) process,

$$(1 - \phi_1 B)(1 - B)X_t = Z_t,$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$

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Now let $Y_t = (1 - B)X_t = X_t - X_{t-1}$, after some rearrangements we have

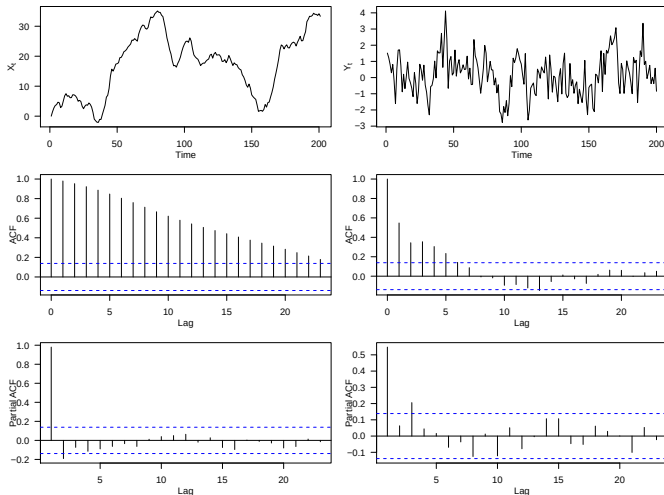
$$\begin{aligned} X_t &= X_{t-1} + Y_t \\ &= (X_{t-2} + Y_{t-1}) + Y_t \\ &\vdots \\ &= X_0 + \sum_{j=1}^t Y_j \end{aligned}$$

Thus $\{X_t\}$ is a “sort of random walk”—we **cumulatively sum** an AR(1) process, $\{Y_t\}$

Simulated ARIMA and Differenced ARMA Process

We simulate an ARIMA(1, 1, 0):

$$(1 - 0.5B)(1 - B)X_t = Z_t, \quad \{Z_t\} \sim N(0, 1)$$



Why Differencing Removes Polynomial Trends

Suppose $X_t \sim \text{ARIMA}(p, d, q)$, so

$$\phi(B)(1 - B)^d X_t = \theta(B)Z_t.$$

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Now add a deterministic polynomial trend:

$$V_t = \mu_t + X_t, \quad \mu_t = a_0 + a_1 t + \cdots + a_{d-1} t^{d-1}.$$

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Key fact

A d th difference eliminates any polynomial of degree less than d :

$$(1 - B)^d \mu_t = 0.$$

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Therefore,

$$\begin{aligned} \phi(B)(1 - B)^d V_t &= \phi(B)(1 - B)^d (\mu_t + X_t) \\ &= \underbrace{\phi(B)(1 - B)^d \mu_t}_0 + \phi(B)(1 - B)^d X_t \\ &= \theta(B)Z_t. \end{aligned}$$

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Takeaway: An ARIMA model with d differences is unchanged by adding a polynomial trend of degree less than d .

Building an ARIMA Model

1. How do we obtain stationarity?

Transform if needed $\longrightarrow$ choose d

2. How do we model the remaining dependence?

ACF/PACF $\longrightarrow$ candidate (p, q) $\longrightarrow$ fit candidate models

3. Is the fitted model adequate?

Residual diagnostics $\longrightarrow$ model comparison $\longrightarrow$ forecast

Residual plots + residual ACF + Ljung-Box AIC/AICc

Key distinction

d : addresses nonstationarity	p, q : model dependence after stationarity
---------------------------------	--

Forecasting ARIMA Processes

We need more assumptions to forecast $\text{ARIMA}(p, d, q)$ processes.
Let us start with the case of $d = 1$, i.e.,

$$\phi(B)(1 - B)X_t = \theta(B)Z_t,$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$

- ▶ **Note:** $Y_t = (1 - B)X_t = X_t - X_{t-1}$ is an $\text{ARMA}(p, q)$ process
- ▶ We want to find the **best linear predictor** (BLP) of X_{n+1} based on $X_0, X_1, \dots, X_n$
 - ▶ We know that $X_{n+1} = X_n + Y_{n+1} \Rightarrow$ only need to figure out the BLP of Y_{n+1} based on $\{X_0, Y_1, \dots, Y_n\}$
 - ▶ We need to know $\mathbb{E}(X_0^2)$ and $\mathbb{E}(X_0 Y_j)$ for $j = 1, \dots, n + 1$

Forecasting ARIMA($p, 1, q$) Processes (Cont'd)

Problem: What is $\mathbb{E}(X_0 Y_j)$?

- ▶ We **assume** that X_0 is **uncorrelated** with $Y_1, Y_2, \dots$

Forecasting ARIMA($p, 1, q$) Processes (Cont'd)

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- ▶ Then the BLP of X_{n+1} based on $\{X_0, X_1, \dots, X_n\}$ is the same as the BLP of X_{n+1} based on $\{Y_1, Y_2, \dots, Y_n\}$

Forecasting ARIMA($p, 1, q$) Processes (Cont'd)

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- ▶ This extends to ARIMA(p, d, q) processes:

If we **assume** that $\{X_{1-d}, \dots, X_0\}$ is **uncorrelated** with $Y_1, Y_2, \dots$, then the BLP of Y_{n+1} based on $\{X_{1-d}, \dots, X_0, \dots, X_n\}$ is the same as the BLP based on $\{Y_1, Y_2, \dots, Y_n\}$

Percentage Changes and Logarithms

Suppose X_t tends to have relatively stable **percentage changes** from one time period to the next. Specifically, assume that

$$X_t = (1 + Y_t)X_{t-1},$$

where $100Y_t$ is the percentage change from X_{t-1} to X_t . Then

$$\log(X_t) - \log(X_{t-1}) = \log\left(\frac{X_t}{X_{t-1}}\right) = \log(1 + Y_t).$$

If Y_t is restricted to, say, $|Y_t| < 0.2$ (i.e., the percentage changes are at most $\pm 20\%$), then, to a good approximation, $\log(1 + Y_t) \approx Y_t$. Consequently

$$\Delta[\log(X_t)] \approx Y_t$$

will be relatively stable and perhaps well-modeled by a stationary process.

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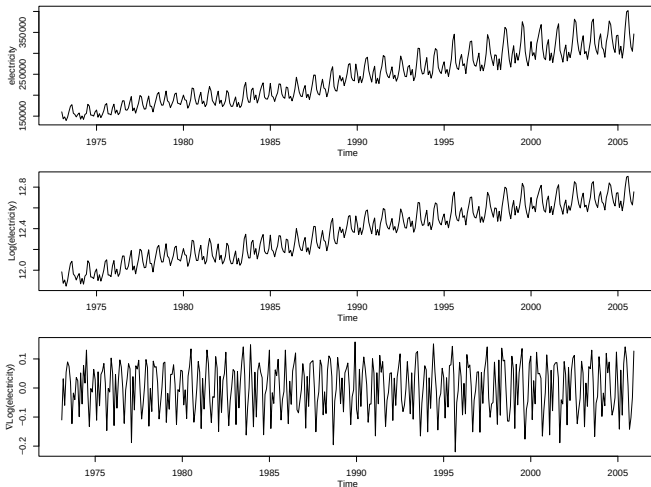
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In the **financial** literature, the differences of the (natural) logarithms are usually called **returns**

Electricity Production: From Nonstationary to Stationary



Step 1: Stabilize the variance **Step 2: Obtain stationarity**

$$X_t \longrightarrow \log(X_t) \qquad \log(X_t) \longrightarrow \text{difference}$$

Seasonal ARIMA (SARIMA) Model

Modeling Trend, Seasonality, and Noise

Recall the trend, seasonality, noise decomposition:

$$Y_t = \mu_t + s_t + \eta_t,$$

where

- ▶ μ_t : (deterministic) trend component;
- ▶ s_t : (deterministic) seasonal component with mean 0;
- ▶ η_t : random noise with $\mathbb{E}(\eta_t) = 0$

We have already described ways to estimate these components. But in many series, the seasonal pattern is not exactly repeated from year to year.

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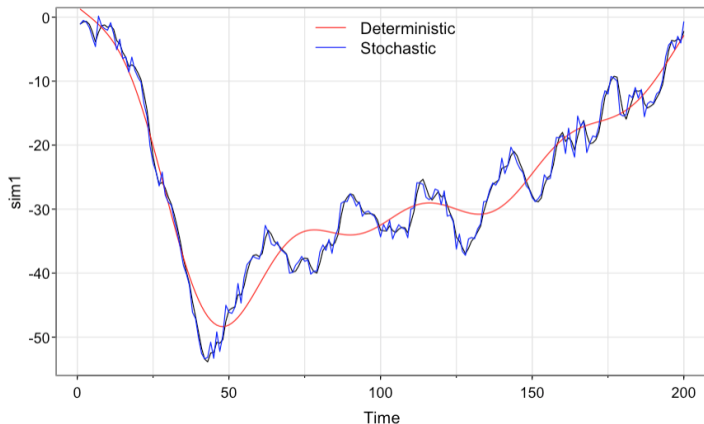
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⇒ The **seasonal ARIMA** model treats seasonal dependence as part of the stochastic time-series structure rather than as a fixed deterministic seasonal curve.

Digression: Using ARIMA for Stochastic Trend Modeling



For a given time series, it may be challenging to identify the exact form of a deterministic trend μ_t . However, **ARIMA** models can effectively capture and account for a “stochastic” trend

Formal Definition of a SARIMA Model

Let d and D be non-negative integers. Then $\{X_t\}$ is a seasonal $\text{ARIMA}(p, d, q) \times (P, D, Q)$ process with period s if

$$Y_t = \nabla^d \nabla_s^D X_t = (1 - B)^d (1 - B^s)^D X_t,$$

is an ARMA process define by

$$\phi(B)\Phi(B^s)Y_t = \theta(B)\Theta(B^s)Z_t,$$

where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$.

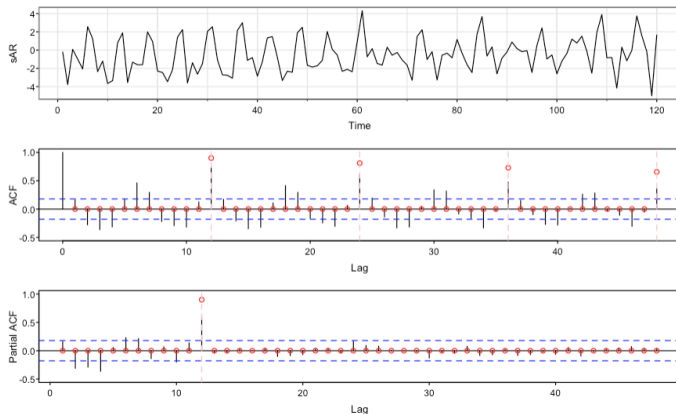
Interpretation

SARIMA extends ARMA by incorporating ordinary and seasonal differencing, together with nonseasonal and seasonal AR and MA components

An Example of a Seasonal AR Model

$$Y_t = 0.9Y_{t-12} + Z_t,$$

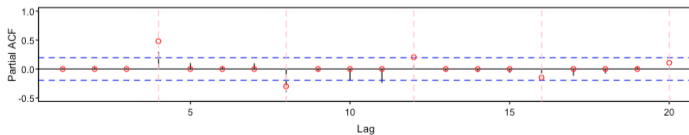
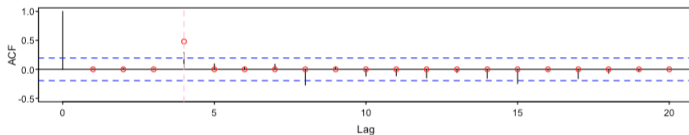
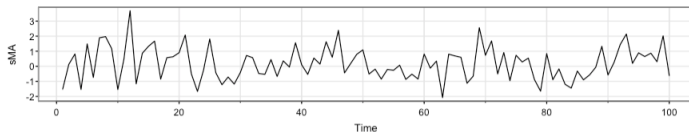
$$\Rightarrow p = q = d = D = Q = 0, P = 1, \Phi_1 = 0.9, s = 12.$$



An Example of a Seasonal MA Model

$$Y_t = Z_t + 0.75Z_{t-4},$$

$$\Rightarrow p = q = d = D = P = 0, Q = 1, \Theta_1 = 0.75, s = 4.$$



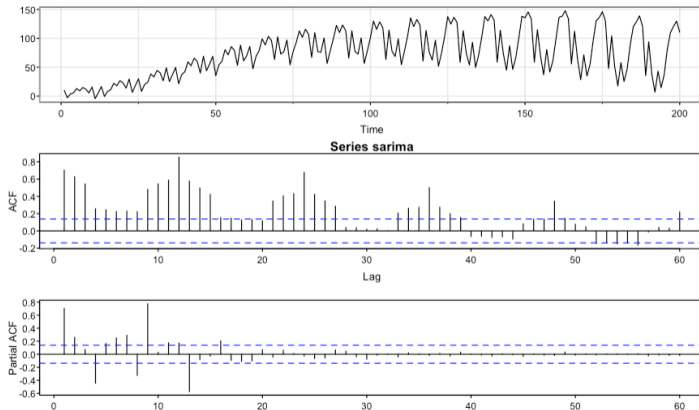
Example of a SARIMA Model

$$(1 - B)(1 - B^{12})X_t = Y_t$$

$$(1 + 0.25B)(1 - 0.9B^{12})Y_t = (1 + 0.75B^{12})Z_t$$

$$\Rightarrow p = P = Q = d = D = 1,$$

$$\phi = -0.25, \Phi = 0.9, \Theta_1 = 0.75, s = 12.$$



An Illustration of SARIMA Model

Consider a monthly time series $\{X_t\}$ with both a trend, and a seasonal component of period $s = 12$.

- ▶ Suppose we know the values of d and D such that $Y_t = (1 - B)^d(1 - B^{12})^D X_t$ is **stationary**
- ▶ We can arrange the data this way:

	Month 1	Month 2	...	Month 12
Year 1	Y_1	Y_2	...	Y_{12}
Year 2	Y_{13}	Y_{14}	...	Y_{24}
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
Year r	$Y_{1+12(r-1)}$	$Y_{2+12(r-1)}$	...	$Y_{12+12(r-1)}$

The Inter-annual Model

Here we view each column (month) of the data table from the previous slide as a **separate time series**

- For each month m , we assume the same $\text{ARMA}(P, Q)$ model. We have

$$\begin{aligned} Y_{m+12s} - \sum_{i=1}^P \Phi_i Y_{m+12(s-i)} \\ = U_{m+12s} + \sum_{j=1}^Q \Phi_j U_{m+12(s-j)}, \end{aligned}$$

for each year $s = 0, \dots, r-1$, where

$\{U_{m+12s:s=0,\dots,r-1}\} \sim \text{WN}(0, \sigma_U^2)$ for each m

- We can write this as

$$\Phi(B^{12})Y_t = \Theta(B^{12})U_t,$$

and this defines the **inter-annual model**

The Intra-Annual Model

After removing the trend, seasonality, and inter-annual effects, the series may still exhibit short-term dependence across months

We model the remaining dependence using an $\text{ARMA}(p, q)$ process:

$$\phi(B)U_t = \theta(B)Z_t,$$

where

$$Z_t \sim WN(0, \sigma^2).$$

SARIMA Key Idea

Annual and monthly differencing remove trends, the annual ARMA model captures inter-year dependence, and the ARMA model captures the remaining month-to-month correlation

Steps for Modeling SARIMA Processes

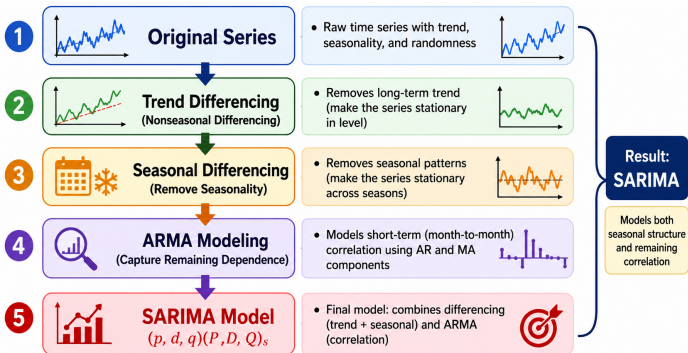
1. Transform the data if necessary (e.g., logarithm for variance stabilization)
2. Apply regular and seasonal differencing until the series is approximately stationary

$$Y_t = (1 - B)^d (1 - B^s)^D X_t$$

3. Use the ACF/PACF at seasonal lags $(s, 2s, 3s, \dots)$ to identify P and Q
4. Use the ACF/PACF at nonseasonal lags $(1, \dots, s - 1)$ to identify p and q

Modeling SARIMA Processes (Cont'd)

5. Use **maximum likelihood method** to fit the models
6. Use model summaries, diagnostics, AIC to determine the best SARIMA model
7. Conduct forecast



Two Types of Dependence in SARIMA

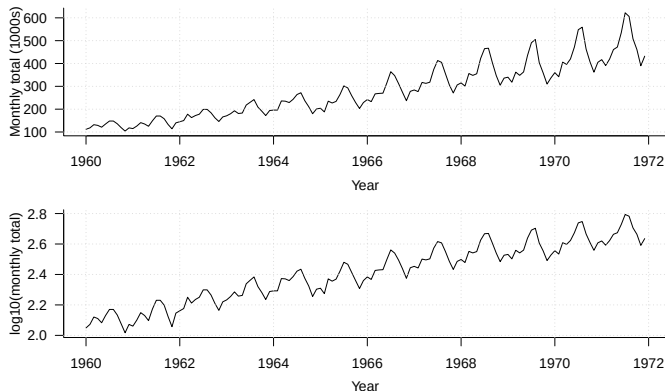
Component	Captures	Example Lag
$AR(p)$	Month-to-month dependence	1,2,3
$MA(q)$	Short-term shocks	1,2,3
$SAR(P)$	Seasonal dependence	12,24,36
$SMA(Q)$	Seasonal shocks	12,24,36

Interpretation

SARIMA combines short-term dependence and seasonal dependence in a single model

Airline Passengers Example [Box and Jenkins, 1970]

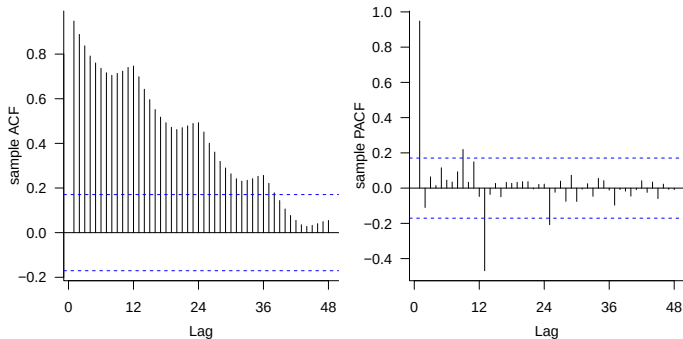
The AirPassengers data record monthly international airline passenger totals from 1949–1960



The seasonal fluctuations increase as the number of passengers grows, indicating nonconstant variance.

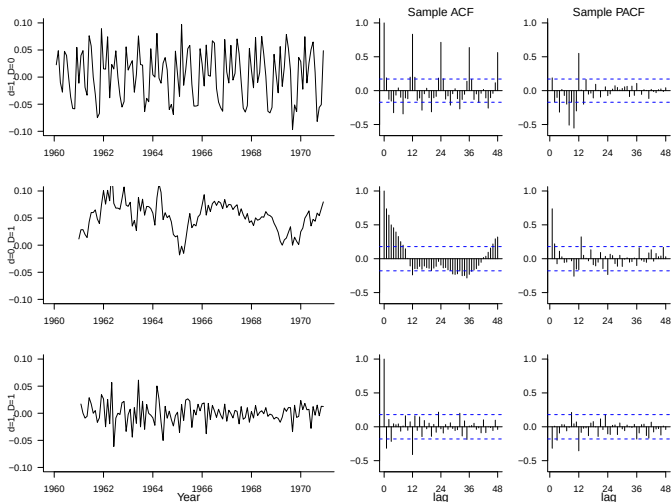
A logarithmic transformation is therefore appropriate

Sample ACF/PACF Plots



- ▶ A slowly decaying ACF and a PACF spike near one at lag 1 suggest nonstationarity and the need for regular differencing
- ▶ Strong spikes at seasonal lags (12, 24, 36) indicate annual seasonality and suggest seasonal differencing

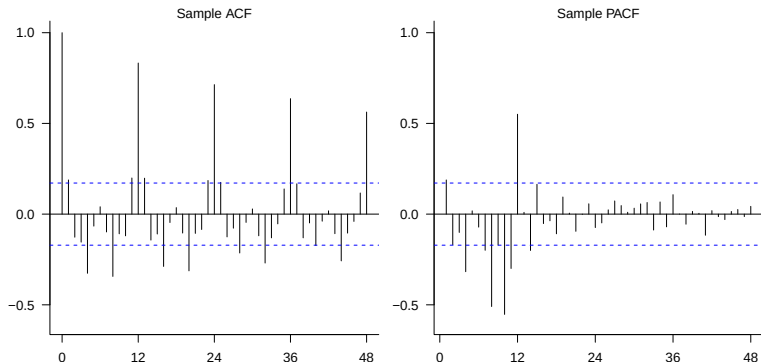
Trying Different Orders of Differencing



Apply the smallest amount of differencing needed to remove trend

Choosing Candidate SARIMA Models

The first difference $d = 1$ appears sufficient to remove the trend. The ACF/PACF will now be used to identify the seasonal and nonseasonal AR/MA orders



Now we need to choose P , Q , p , and q

Fitting a SARIMA(1, 1, 0) $\times$ (1, 0, 0) Model

```
> fit1 <- arima(diff.1.0, order = c(1, 0, 0), seasonal = list(order = c(1, 0, 0), period = 12))  
> fit1
```

Call:

```
arima(x = diff.1.0, order = c(1, 0, 0), seasonal = list(order = c(1, 0, 0),  
  period = 12))
```

Coefficients:

	ar1	sar1	intercept
	-0.2667	0.9291	0.0039
s.e.	0.0865	0.0235	0.0096

sigma^2 estimated as 0.0003298: log likelihood = 327.27, aic = -646.54

Estimated Model

Both the nonseasonal AR(1) and seasonal AR(1) terms are statistically significant.

Diagnostic Check: SARIMA(1, 1, 0) $\times$ (1, 0, 0)₁₂

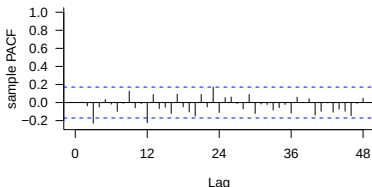
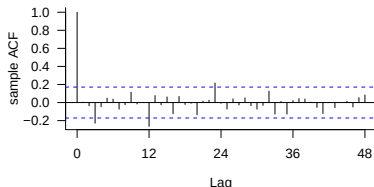
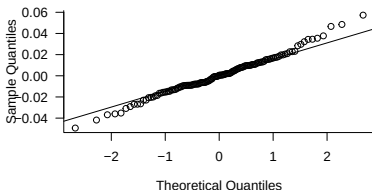
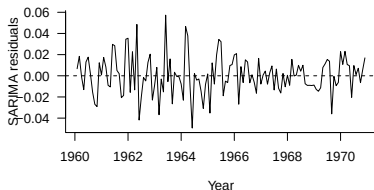
```
> Box.test(fit1$residuals, lag = 48, type = "Ljung-Box")
```

Box-Ljung test

data: fit1\$residuals

X-squared = 55.372, df = 48, p-value = 0.2164

Ljung-Box test examines residual independence



A Discussion of the Model Fit

- ▶ The spread of the residuals is larger in 1949-1955 compared to the later years and the residual distribution has heavy tails
- ▶ The Ljung-Box test result indicates the fitted SARIMA $(1, 1, 0) \times (1, 0, 0)$ has sufficiently account for the temporal dependence
- ▶ 95% CI for ϕ_1 and Φ_1 do not contain zero $\Rightarrow$ no need to go with simpler model

Our estimated model is

$$(1 + 0.2667B)(1 - 0.9291B^{12})(X_t - 0.0039) = Z_t,$$

where $\{Z_t\} \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$ with $\hat{\sigma}^2 = 0.00033$

Comparing with a SARIMA(0, 1, 0) $\times$ (1, 0, 0) Model

```
> (fit2 <- arima(diff.1.0, seasonal = list(order = c(1, 0, 0), period = 12)))
```

Call:

```
arima(x = diff.1.0, seasonal = list(order = c(1, 0, 0), period = 12))
```

Coefficients:

	sar1	intercept
	0.9081	0.0040
s.e.	0.0278	0.0108

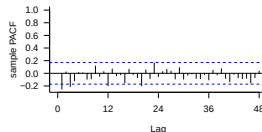
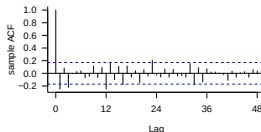
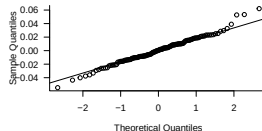
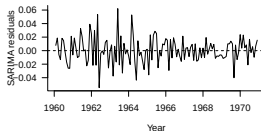
sigma^2 estimated as 0.0003616: log likelihood = 322.75, aic = -639.51

```
> Box.test(fit2$residuals, lag = 48, type = "Ljung-Box")
```

Box-Ljung test

data: fit2\$residuals

X-squared = 80.641, df = 48, p-value = 0.002209



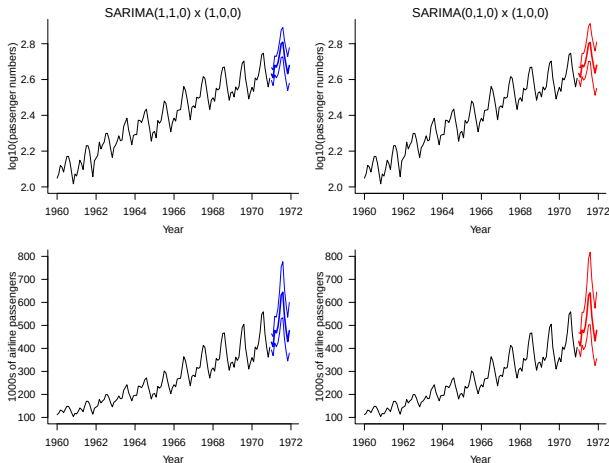
A Discussion of Model Fit2

Here we drop the AR(1) term

- ▶ The residual plots looks quite similar to before: The spread of the residuals is larger in 1949-1955 compared to the later years and the residual distribution has heavy tails
- ▶ Both $\hat{\sigma}^2$ and AIC increase (compared with model fit1)
- ▶ The lag 1 of ACF and PACF now lies outside the IID noise bounds. The Ljung-Box p -value of 0.0022, leads us to reject the IID residual assumption

In conclusion, the $\text{SARIMA}(1, 1, 0) \times (1, 0, 0)$ model fits better than $\text{SARIMA}(0, 1, 0) \times (1, 0, 0)$

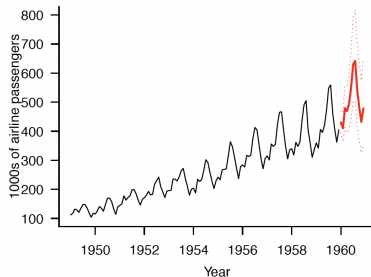
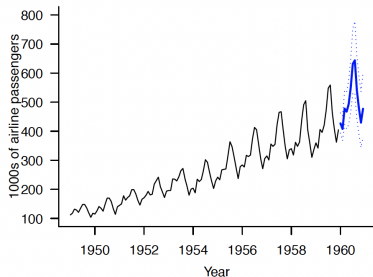
Forecasting the 1960 Data



Good forecasting performance requires

1. Accurate point forecasts
2. Reliable prediction intervals

Evaluating Forecast Performance



Metrics	Model Fit1	Model Fit2
Root Mean Square Error	30.36	31.32
Mean Relative Error	0.057	0.060
Empirical Coverage	0.917	1.000

The SARIMA(1, 1, 0) $\times$ (1, 0, 0) Model is Equivalent To?

Our model for the log passenger series $\{X_t\}$ is

$$\phi(B)\Phi(B^{12})(1-B)X_t = Z_t,$$

where $\phi(B) = 1 - \phi_1 B$ and $\Phi(B) = 1 - \Phi_1(B)$

Note that

$$\begin{aligned}\phi(B)\Phi(B^{12}) &= (1 - \phi_1 B)(1 - \Phi_1 B^{12}) \\ &= 1 - \phi_1 B - \Phi_1 B^{12} + \phi_1 \Phi_1 B^{13}\end{aligned}$$

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Question: What is this SARIMA model equivalent to?